querying and processing of large datasets to support reporting and analysis, transactional data stores optimize for fast, consistent data input and modification.

- Relational databases remain essential for reference data, asset hierarchies, and operational records that benefit from normalized schemas and referential integrity. Modern cloud databases provide managed services that reduce operational overhead while maintaining SQL compatibility.
- NoSQL databases emerged to address limitations of relational databases when handling large-scale, diverse data with flexible schemas and horizontal scaling requirements. Digital twins often work with just this type of data from high-volume sensor streams, varying data structures across device types, and unpredictable scaling demands as systems grow from pilot projects to enterprise deployments.

4.5.5 Specialized storage systems

Digital twins often require specialized storage capabilities for specific data types and use cases:

- *Object storage* provides scalable, cost-effective storage for unstructured data including images, videos, documents, and large files. Cloud object storage services offer multiple storage tiers, automatic lifecycle management, and global distribution capabilities.
- *Graph databases* excel at storing and querying complex relationships between entities. These systems use specialized algorithms for traversing connections and identifying patterns across interconnected data.
- *Vector databases* enable semantic similarity searches across high-dimensional data by storing embeddings and providing optimized nearest-neighbor search algorithms. These systems support applications like document similarity, anomaly detection, and recommendation engines.
- *Feature stores* centralize engineered features for machine learning, ensuring consistency between training and serving environments while providing versioning, lineage tracking, and low-latency access for real-time predictions.

4.5.6 Data lifecycle management

The age of the data you collect and store also influences the storage technology that you choose, and data may transition through different types of storage over time. Recent data typically needs fast access for operational decision making, including alerts and dashboard visualization; whereas older data, while still important for analytical purposes, can often tolerate slower query times.